6.1 Group 2 (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	6. Inorganic Chemistry (A Level Only)
Topic	6.1 Group 2 (A Level Only)
Difficulty	Easy

Time allowed: 30

Score: /22

Percentage: /100

Question 1a

This question is about Group 2 carbonates.

Magnesium carbonate is used as an antacid and used to treat heartburn.

i)
Write the chemical formula of magnesium carbonate.

[1]

ii)
Write a balanced symbol equation for the reaction of magnesium carbonate with dilute nitric acid.

[2]

[3 marks]

Question 1b

i)
From the list below, identify the compound that will decompose at the lowest temperature.

- calcium carbonate
- strontium carbonate
- barium carbonate

[1]

ii)
Write a balanced symbol equation for the decomposition of your chosen carbonate.

[1]

[2 marks]

Question 1c

Group 2 nitrates become more thermally stable going down the group.

Explain why.

[2 marks]

Question 2a

This question is about Group 2 compounds.

Strontium will react with water.

i)
Write a balanced symbol equation for the reaction.

[2]

ii)
Describe a test to identify the gas produced.

[1]

[3 marks]

Question 2b

Calcium hydroxide reacts with sulfuric acid to produce calcium sulfate and water.

i)
Write the balanced symbol equation for this reaction.

[2]

ii)
Describe the trend in the solubility of the Group 2 sulfates.

[1]

[3 marks]

Question 2c

The trend in solubility of the Group 2 sulfates is due to the lattice energy, ΔH^{θ}_{latt}, and enthalpy change of hydration, ΔH^{θ}_{hyd} , of the Group 2 sulfates changing.

Describe the trend in ΔH^{θ}_{latt} and ΔH^{θ}_{hyd} of Group 2 sulfates going down Group 2.

[2 marks]

Question 3a

The elements in Group 2 from Mg to Ba can be used to show the trends in properties down a group in the Periodic Table.

The Group 2 elements react with water.

i)
State the trend in reactivity with water of the elements down Group 2 from Mg to Ba.

[1]

ii)
Predict a possible pH for the solutions formed when Group 2 elements are added to water.

[1]

[2 marks]

Question 3b

Give the **formula** of the hydroxide of the element in Group 2 from Mg to Ba that is most soluble in water.

[1 mark]

Question 3c

A trend in the thermal decomposition of Group 2 nitrates can also be observed.

i)
Give the oxidation number of nitrogen in $Ca(NO_3)_2$.

[1]

ii)
Write an equation for the thermal decomposition of $Ca(NO_3)_2$.

[2]

iii)
State the trend in thermal stability of Group 2 nitrates.

[1]

[4 marks]

Model Answers: Easy

1a

a)

i)The chemical formula of magnesium carbonate is:

- $MgCO_3$; [1 mark]

ii) A balanced symbol equation for the reaction of magnesium carbonate with dilute nitric acid is:

$$MgCO_3(s) + 2HNO_3 \rightarrow Mg(NO_3)_2 + CO_2 + H_2O$$

- Correct formulae; [1 mark]
- Correct balancing; [1 mark]

[Total: 3 marks]

- The general equation for the reaction of a metal carbonate and acid is:
  - Metal carbonate + acid $\rightarrow$ salt + carbon dioxide + water
- You are expected to be able to deduce the formulae for each substance involved

1b

b)

i) The compound that will decompose at the lowest temperature is:

- Calcium carbonate; [1 mark]

ii) A balanced symbol equation for the decomposition for this carbonate is:

- $CaCO_3$ (s) $\rightarrow$ CaO (s) + CO_2 (s); [1 mark]

[Total: 2 marks]

- Going down Group 2, more heat is needed to break down carbonates so the one that will require the lowest temperature is the one containing a Group 2 metals from higher up Group 2
- The symbol equation for the thermal decomposition of each Group 2 carbonate is the same, make sure you include state symbols
- If you identified a different Group 2 carbonate for your answer to part i) you can still score the mark for part ii) if you have written the symbol equation correctly

1c

c) Group 2 nitrates become more thermally stable going down the group because:

- The size / radius of the cation increases
  OR
  Charge density of the ion decreases; [1 mark]
- So polarisation / distortion of anion / nitrate ion decreases; [1 mark]

[Total: 2 marks]

- At A level you must be able to explain the trends in the thermal stabilities of Group 2 carbonates and nitrates down the group
- The explanation for both is the same:
  - As the size of the Group 2 ion increases, it's polarising power decreases
  - A decrease in the polarisation of the nitrate / carbonate ions means it is harder to weaken the N-O / C-O bond so the compound does not decompose easily

2a

a)

i) A balanced symbol equation for the reaction is:

- $Sr (s) + 2H_2O (l) \rightarrow Sr(OH)_2 (s) + H_2 (g)$; [1 mark]

ii) A test to identify the gas produced is:

- Burning splint makes a (squeaky) pop sound; [1 mark]

[Total: 2 marks]

- You must have the correct state symbols to score the mark in part i)
- A common mistake is to identify $Sr(OH)_2$ as aqueous
- Both the test and result for hydrogen gas must be present in part ii) to score 1 mark

2b

c)

i) A balanced symbol equation for this reaction is:

- $Ca(OH)_2 + H_2SO_4 \rightarrow CaSO_4 + 2H_2O$; [1 mark]

ii) The trend in the solubility of the Group 2 sulfates is:

- Going down Group 2, solubility of the Group 2 sulfates decrease; [1 mark]

[Total: 2 marks]

- The general equation for the reaction in part i) is)
  - Metal hydroxide + acid → salt + water
- Going down the solubility of Group 2 sulfates decreases but for Group 2 hydroxides it increases
- For harder questions you can be expected to explain these trends

2c

c) The trend in ΔH^{θ}_{latt} and ΔH^{θ}_{hyd} of Group 2 sulfates going down Group 2 is:

- Lattice energy decreases; [1 mark]
- Enthalpy change of hydration decreases; [1 mark]

[Total: 2 marks]

- **Careful**: Both lattice energy and enthalpy change of hydration decrease but by different amounts
  - The lattice energy of the Group 2 sulfates decreases by a relatively small amount going down the group
  - The enthalpy change of hydration decreases by a a relatively large amount going down the group because the size of the ions increases
- As a result of these trends, the value for ΔH^{θ}_{sol} becomes more endothermic

going down the group and solubility decreases

3a

a)

i) The trend in reactivity with water of the elements down Group 2 from Mg to Ba is:

- (Reactivity)increases; [1 mark]

ii) A possible pH for the solutions formed when Group 2 elements are added to water is:

- 10–12; [1 mark]

[Total: 2 marks]

- Similar to Group 1, Group 2 elements react more vigorously the further down the group you go
  - There is no reaction between berylium and water and magnesium will only react with steam
- A metal hydroxide is formed, containing the OH^- ion making the resulting solution alkaline
- These solutions usually have a pH in the range of 10–12

3b

b) The **formula** of the hydroxide of the element in Group 2 from Mg to Ba that is most soluble in water is:

- $Ba(OH)_2$; [1 mark]

[Total: 1]

- The solubility of Group 2 hydroxides increases going down the group
- Magnesium hydroxide is only sparingly soluble

3c

c)

i) The oxidation number of nitrogen in $Ca(NO_3)_2$ is:

- +5; [1 mark]

ii) The equation for the thermal decomposition of $Ca(NO_3)_2$ is:

- $2Ca(NO_3)_2 (s) \rightarrow 2CaO (s) + 4NO_2 (g) + O_2 (g)$; [1 mark]

iii) The trend in thermal stability of Group 2 nitrates is:

- Going down Group 2, the thermal stability increases
  OR
  Group 2 nitrates are more thermally stable going down the group; [1 mark]

[Total: 3 marks]

- The sum of the oxidation numbers for the compound should equal 0
  - The oxidation number of oxygen is -2, of which there are three, so -12
  - The oxidation number of calcium is +2 so -12 + 2 = -10
  - There are two nitrogen atoms so each one must have an oxidation number of +5 to be 0 overall